

Solution Presentation

Solution Presentation - Sentinel AI (Gujarat CCTV Integration)

> Deliverable 1 as per hackathon guidelines. Markdown outline for PPT/PDF export (Marp / PowerPoint). Each heading = one slide; speaker notes included as blockquotes.

Slide 1 - Title

> Speaker note: One platform that onboards 50 heterogeneous feeds today and scales to statewide 80k without redesign.

Slide 2 - Proposed Solution Model & Justification

| Option evaluated | Why hybrid 3+4 was chosen |

> Architecture principles compliance: open standards (ONVIF/RTSP/HLS/WebRTC/REST), adapter-based vendor neutrality, additive migrations, technology-agnostic - see `05\_Architecture\_Principles\_Compliance.md`.

Slide 3 - Solution Overview & Objectives

Objectives (from portal): integrate ~50 heterogeneous feeds, correlate live video with watchlist DB, AI-powered real-time alerts.

Overview:

Slide 4 - Key Innovations

1. **AdapterRegistry** - one `CameraAdapter` contract per vendor (Hikvision/Dahua/Axis/Bosch/Hanwha/CP Plus/ONVIF); `register()` adds a brand without core change.

Slide 5 - System Architecture Diagram

flowchart LR

> Data flow: ingest -> registry (ownership/GIS) -> ANPR -> watchlist match -> alert -> dossier/interception -> cluster HA -> web.

Slide 6 - Camera Onboarding for ~50 Heterogeneous Cameras

| Mechanism | How it works | Evidence |

Onboarded fleet: `GET /api/v1/cameras` (51), `GET /api/v1/registry` (51), `GET /api/v1/registry/gis/report`.

Slide 7 - AI & Video Analytics Approach

ANPR: `src/anpr/` (primitives, tracker `norFair`-style, backend `sim`), OCR conf 0.86-0.93, `backend="sim"` swappable to live engine behind same `anpr/` contract; benchmark harness `media/ai/benchmarks/` (sub-second).

Watchlist matching: `normalized\_plate` lookup against `Watchlist (identifier\_number, source\_db)`; demonstration chains seeded for `GJ01AB1234` (5 hops AHM) + `GJ06XY9012` (3 hops SRT).

Alerting: `BLACKLIST` on first hit, `MULTI\_CAMERA` on 2nd distinct camera within window - both fired live (`demo\_scenario Act III [PASS]`). Alerts surface at `GET /alerts`, `GET /anpr/alerts`, `app/alerts`, WebSocket `inference/ws`.

Dossier + Interception: `src/forensics/dossier.py` (manifest SHA), `src/anpr/interception.py` (corridor).

Slide 8 - Scalability Roadmap to ~80,000 Cameras

| Tier | Today (51) | 80k design |

> Proof: `SCALABILITY.md` + `05` §2.4 + `api/v1/cluster/` (register/heartbeat/failover) tested; GIS gap engine runs over 10800 cells on seed.

Slide 9 - Network & Storage Considerations

- ****Ingest:**** RTSP over district MPLS -> state DC; MediaMTX remux `hlsAlwaysRemux=yes`, `hlsSegmentCount=15`, `1s` segments.

Slide 10 - Cost-Benefit & Resource Estimation (per 80k)

- Vendor-neutral saves ~30-40% vs single-VMS lock-in (replace per-vendor without forklift).

> Full numbers in `02\_Vendor\_Evaluation\_Criteria.md` + `04\_RFP\_Document.md`.

Slide 11 - Live Evidence (Screenshots to record for video)

- `GET /vehicles/GJ01AB1234/dossier` -> 5 sightings route (AHM chain 17:53->18:31 UTC)

Slide 12 - Deployment & Submission Links

- `docker compose up --build -d` -> `http://localhost:8000/docs` (OpenAPI), `http://localhost:3000/login` (admin@sentinel.gp / Admin@2026)